From Upstream to Downstream: Cascading Risks in the Jeneberang Watershed, South Sulawesi, Indonesia

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Abstract The Jeneberang watershed is one of degraded watersheds and is included in the 15 national priority watersheds in Indonesia. By using an ecosystem approach, this study aims to analyze the deforestation in Jeneberang that pose a cascading risks to the ecosystem sustainability throughout the watershed, starting from the upstream, middle-stream to downstream areas. The methods consist of a literature review, spatial analysis, focus group discussions (FGD), in-depth interviews, and field observations. This study reveals that human activities dominate deforestation along the Jeneberang watershed with activities varying by watershed areas. The forest cover in the upstream area is only 16.8% of 78,883.9 hectares. The deforestation is mainly caused by horticultural agriculture and tourism development. Meanwhile, in the middle and downstream areas, massive land-use changes and technological interventions occur through infrastructure development such as Bili-Bili Reservoir, dykes, and sand pockets. These activities coupled with physiography of Jeneberang River and extreme weather conditions which have a high impact on cascading risks, including landslides in the upstream which affect high sedimentation in the Jeneberang River and Bili-Bili Reservoir as well as flash floods during the rainy season and the risk of drought in the dry season. The cascading risks in Jeneberang Watershed also affects the decline in quantity and quality of water in the river and Reservoir leading to a lack of water sources for irrigating rice fields and a shortage of clean and drinking water for residents in the downstream and urban areas including Makassar City.

Keywords Deforestation · Disaster · Cascading risks · Jeneberang watershed

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1 Introduction

Exploitation of natural resources, especially forests and sand mining in Jeneberang Watershed, is practically a necessity for the life and livelihood of the community and development in South Sulawesi and particularly in the surrounding area. Forest conversion was initially seen as the impact of population growth, especially farmers in forest areas, turned into a massive exploitation activity after many actors played a bigger role in the use of forests for economic development and commercial purposes. Forest exploitation is no longer carried out merely to meet the needs of the communities, but for the development of various sectors in many cities, such as the tourism, mining, industry, and infrastructure development as well as various city facilities. This exploitation is increasing with the increase in economic and development activities which generally rely on natural resources.

In many regions in the world, deforestation has become one of anthropogenic activities that destructively degrades biodiversity and environmental services which is actually very important for humans [8]. According to Meyer et al. [20], the simple definition of deforestation is the conversion of forest areas to become non-forest areas, usually for agriculture activity, settlements, or public infrastructures such as dams or roads. In short, deforestation will reduce the strategic function of forest areas and will be more complicated and more difficult to control when the human population is still increasing [13]. Moreover, the risk of deforestation could increase in recent times because of technological development that is more advanced to extract forest resources than the last time.

In the context of Jeneberang Watershed, the forest in the upstream is protected by law through the Government Regulation no. 37/2012 and the people especially those who live in the upstream area [21]. Unfortunately, the Jeneberang Watershed in fact has a worrisome condition that relates to forest covers, because majority of forest cover (83.18%) in Jeneberang Watershed were converted to horticultural crops (41.20%), paddy field (28.29%), and settlements (8.86%) [32]. Despite knowing that the forest is part of a conservation area, the authorities tend to ignore it so that deforestation becomes more massive and uncontrolled. Moreover, there are some agrarian issues that complicate the situation, such as land certificates issues which emerged within communities living near protected forest areas.

Exploitation of natural resources which is full of economic interests and financial benefits often leads to environmental destruction and thus affects people's vulnerabilities in many regions. Excessive consumption of natural resources results in environmental degradation due to reduced effectiveness of important ecosystem services, such as disruption of infiltration and water discharge which causes soil erosion, sedimentation, flooding in the rainy season and drought in the dry season. This in turn leads to a *cascading risks*, a condition where there is an increase in natural hazards that can further damage the environment and risk the communities [1, 9, 10].

Disaster risk reduction approaches are becoming a challenge in today's world as hazards become more frequent and more complex. This is fueled by population and economic growth, increasing interrelated exposures leading to more systemic

vulnerabilities. The quantification and prediction of risk is therefore not easy, mainly because of the complexities underlying cascading risks.

Cascading risks is a risk that is endogenous, does not stand alone, but has the potential for latent or cumulative risks that can have a negative impact on the system, occurring when some characteristics of the system change. Cascading risks events may occur suddenly or may increase over time, especially if the response to signal changes is inappropriate. The degree of damage inflicted therefore depends on the underlying process and the severity of the triggering event, how extreme the event triggering the risk is [36].

Some cascading risks focus on national-scale risks required for public policy and provide guidance for disaster risk reduction. However, risks are interconnected across geographic areas, with larger and/or smaller scales. Cascading risks affects areas from upstream to downstream or across urban and/or rural areas, due to the unique characteristics of urban/rural areas as part of a complex system. For example, urban areas in coastal zones are at high risk of sea level rise and flood risk especially in areas without adequate structural measures or behavioral adjustments to avoid precipitating events or cascading hazard processes [36].

Within the Sendai Framework for disaster risk reduction, it was argued that understanding of cascading risks is needed to manage multidimensional and interdependent cascading risks with different systems or across different geographic or spatial scales. An understanding of cascading risks therefore must consider the risk from occasional or extraordinary events, such as flash floods, earthquakes, landslides, extreme weather events and sea level rises [36].

In attempt to understand cascading risks, this research uses ecosystem approach to analyze the process of cascading risks and its impact to the communities. Various literatures have explained that the ecosystem approach is part of integrated natural resource management which is defined as the process of combining various aspects of natural resource use into a sustainable management system to meet the objectives of environmental benefits (e.g., profitability, as well as risk reduction) and the goals of the broader community. In the context of this study, the ecosystem is a socio-ecological system that spans across Jeneberang watershed area, consisting of an upstream area, a middle-stream, and a downstream area, from Gowa Regency to Makassar City.

Watershed damage in Indonesia is increasing every year, so that it's causing very detrimental excesses. Every year there are always floods, landslides, droughts and other natural disasters. One of the causes of watershed damage is due to land use planning and management practices that are not in accordance with the right pattern [23]. The level of watershed damage can be determined from several physical indicators, namely the ratio of maximum/minimum river discharge, run-off coefficient, erosion, and sedimentation, ground water level, and spring discharge [27].

In this article, we aim to answer two research questions: 1. How anthropogenic activities especially deforestation in the upstream of Jeneberang watershed lead to a cascading risks within the whole Jeneberang watershed? 2. How is the process of cascading risks and what are the impacts to the communities in South Sulawesi particularly those who live within Jeneberang watershed? Hypothetically, we believe that

deforestation or forest degradation in Jeneberang watershed has caused vulnerabilities that pose a cascading risks to ecosystem sustainability throughout the watershed, starting from the upstream, middle-stream to downstream areas. This critical condition of forest areas in Jeneberang Watershed envisages the emergence of risk disasters that could affect the middle stream and downstream area according to Soewarno in 1991 [12] and Noordwijk et al. [22]. This condition is supported by the position of South Sulawesi Province which is geographically prone to disasters [7]. Thus, the level of vulnerability to disasters may increase with the large number of people living in disaster-risk areas, from the upstream to the downstream areas of the watershed.

2 Methods

This article aims to elaborate on how deforestation or forest degradation in Jeneberang watershed has caused vulnerabilities that pose a cascading risks to ecosystem sustainability throughout South Sulawesi particularly in Jeneberang watershed area. We use ecosystem approach to analyze chronologically the process of cascading risks from upstream to the downstream in Jeneberang Watershed and its impact to the communities. The methods consist of a literature review, spatial analysis, FGD, in-depth interviews, and field observations.

For technical analysis and writing, we divide Jeneberang watershed into three segments, namely: upstream, middle stream, and downstream areas. The Jeneberang watershed has an area of ± 78 thousand hectares, with the Malino Nature Tourism Park (TWA) upstream in Gowa Regency, then crossing through Gowa Regency and ending in the downstream coast of Makassar City and Gowa Regency (Fig. 1).

3 Results and Discussions

In this section, we will explain the physiography of Jeneberang watershed and land degradation before explaining in detail how each watershed segment has created cascading risks in Jeneberang watershed. In attempt to make it chronological, this section consists of five sub-sections: (1) physiography and land degradation, (2) upstream: deforestation, horticulture, and tourism development (3) middle stream: sand mining and sedimentation on Bili-Bili Reservoir (4) downstream: damages and its impacts, (5) cascading risks and challenges.

3.1 Physiography of Jeneberang and Land Degradation

Geologically, the Jeneberang watershed is dominated by volcanic and marine sedimentary rocks in the upstream to middle stream areas [31]. Thus, the upstream area

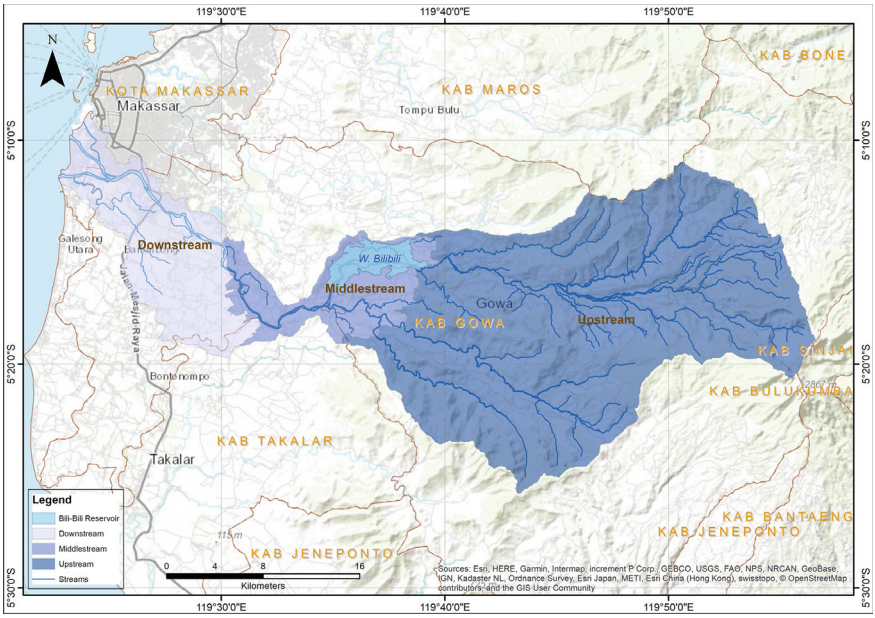

Fig. 1 Study area in Jeneberang Watershed, South Sulawesi

and some part of the middle stream area have fertile soil which is very beneficial to the economy of communities. In terms of geomorphology, recent study has also explained that Jeneberang River is a braided river which is characterized by abundant supply of sediment and high stream gradient [19]. In the upstream, the slope has a tilt degree over 45° that covers 71.1% areas [11]. Jeneberang River also has the highest maximum river discharge among the 20 major rivers in the South Sulawesi region, which is 315.20 m^3/s [18]. Therefore, with the highest river discharge and physical attributes, Jeneberang River is naturally rich in mining materials, such as sand and gravel [19].

Despite having fertile soils and abundant supply of mining materials in the river areas, the land in Jeneberang watershed has been slowly degraded. The Ministry of Environment and Forestry has classified the land degradation level for every region in Indonesia. Based on the map below (Fig. 2), land degradation in Jeneberang watershed is varied but dominated by *potentially degraded, moderately degraded, degraded*, and *highly degraded.* Based on the overlay results, 19.40% (11,314.71 ha) of the upstream area is *highly degraded*. In addition, 47.52% (27,718.43 ha) of the upstream area is in a *potentially degraded* condition. Likewise, what happened in the middle stream, 81.52% (8035.64 ha) of land in the middle region was in a *potentially degraded* condition followed by *degraded* conditions (5.99%) and *highly degraded* (5.81%). Meanwhile, in the downstream area, 89.09% (3428.05 ha) of the area is classified as *non-degraded.* For more details, see the following graph below (Table 1).

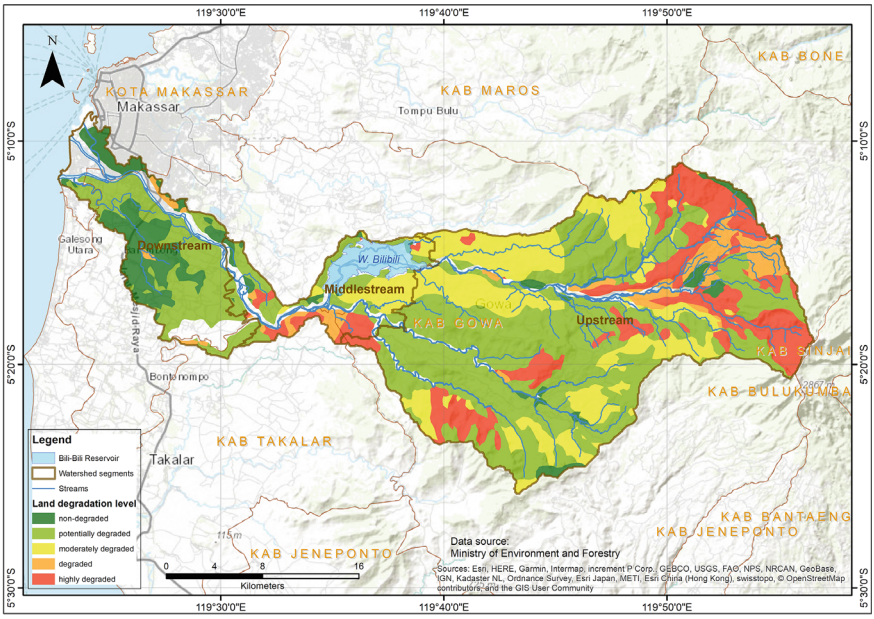

Fig. 2 Distribution of land degradation in Jeneberang watershed

Table 1 Proportion of land degradation in Jeneberang Watershed

Watershed segments	Land degradation level (%)				
	Non-degraded	Potentially degraded	Moderately degraded	Degraded	Highly degraded
Upstream	3.94	47.52	24.26	4.88	19.40
Middle stream	5.51	81.52	1.16	5.99	5.81
Downstream	89.09	3.04	–	7.87	–

3.2 Upstream: Deforestation, Horticulture, and Tourism Development

Based on land degradation data, more than 50% area of the upstream is degraded 35,236 ha (57.08%) indicating that there are quite massive land use changes in the upstream. Human activities such as forest exploitation, tourism development, and the horticultural industry in the upstream have greatly affected the ecological conditions of the Jeneberang watershed and have degraded the soils, making it prone to disasters such as landslides and floods. The land use changes particularly forest conversion caused by several factors which are mostly close to socio-political aspects.

Furthermore, forest conversion in Jeneberang Watershed into horticultural fields by communities has worsen due to the local government's support on vegetable

production in the upstream area. The local government, especially provincial government, have their own interest to boost vegetables production for their regional economic growth. Besides, there is another crucial issue that cause the problem becomes more complex. That is the unclear delineation between conservation/protected areas and communities' arable land. Informant in the upstream stated that the authorities did not really know where exactly the border of conservation areas are. The people said they have been cultivating the land for years, while the forest agency from the central government did not have the document that prove that certain border. This complex and difficult situation could happen because the local government has not signed the conservation borderline which had stated by the central government. Moreover, it had become more complicated because of the land certificate or at least the annual property tax (SPPT-PBB[1]) had been published by the land registry office (BPN). These land certificates or annual property tax (SPPT-PBB) as the retribution from people to local government become the justification for their right to their arable land. These conditions made the government authorities who has same interest to conserve forest areas difficult to battle the encroachment in Jeneberang Watershed.

Massive encroachment in Jeneberang Watershed leave 16.82% of forest areas remained, where approximately 41.2% of the areas are used for horticulture cultivation and 28.3% areas are used as the rice field [32]. In specific data, around 1.688,76 hectares (48.25%) from 3.500 hectares are converted in Malino Nature Park (TWA Malino). While the other source stated that the total area removed outside of Malino Nature Park in Bawakaraeng-Lompobattang's forest area has reached 5.000,34 ha.

In the context of land degradation, there is a dilemma between communities and the government. Historically, before massive encroachment in the upstream of Jeneberang, many people around the forest had depended on the logging that were sold to the Gowa Paper Factory [12]. In the 1980s, horticultural cultivation was brought by farmers from West Java with satisfied output especially in the profit aspect. It later then started to become mainstream until now. The horticultural sector has proved their role in Gowa Regency that they have contributed around 60.9% of agriculture production and 19.6% of Gross Regional Domestic Product [34]. Beside horticultural industries, there is also tourism development in the upstream. The rise of the tourism industry in Malino region could be seen by the sporadic "*villa boom*" or massive construction of villas in the middle of agricultural lands, especially in Batulapisi area. Extensive built-up area will disturb the hydrology system in Jeneberang watershed.

Therefore, deforestation in the upstream of Jeneberang Watershed affects many aspects including water availability in a dry season. The water tends to dry quickly affecting the process of horticultural cultivation production. Some of the farmers tried to stem the water with those who live in lower areas who also need the water for horticultural cultivation. The stem of the upper on the hill by higher upstream people reduces the flow of water for lower upstream people, the flow becomes slow or completely dry in lower upstream areas. They also come up with other strategies, such as trying to dig a well to water their crops. Some of them may have the capacity

[1] Surat Pemberitahuan Pajak Terutang—Pajak Bumi dan Bangunan.

in tackling the water issue when the dry season comes, but some other people may not. So, the water issue in the upstream especially in the dry season could be one of the issues that could cause conflict between the people.

The further effect of forest degradation areas in the upstream of Jeneberang Watershed is the high inequality of water discharge that reaches a ratio of 1: 60 compared with normal watershed water discharge at 1:25 [38]. The ratio is showing that water discharge can be very low when the dry season comes, while in rainy seasons, there will be plenty of water flowing in Jeneberang Watershed. The condition of low water discharge of Jeneberang in the dry season is utilized by sand miners to extract sand directly from riverbeds. On the other hand, the amount of wat.er discharge will be double compared to the water discharge on normal conditions. This higher water discharge in Jeneberang could raise the risk of floods.

3.3 Middle Stream: Sand Mining and Sedimentation on Bili-Bili Reservoir

The main issue in the middle stream area of Jeneberang Watershed is massive sand mining activities and sedimentation due to the poor management of Bili-Bili Reservoir. Based on land degradation data, 81.52% (8035.64 ha) of area in the middle stream is considered as a *potentially degraded* condition followed by *degraded* conditions (5.99%) and *highly degraded* conditions (5.81%). The existence of sand mining including dredging on the banks has caused the river widens and when it rains, the riverbank cannot hold water, water overflows into settlements (Fig. 3).

Fig. 3 Middle stream of Jeneberang

Recent study stated that sand mining activities on the Jeneberang River were first carried out traditionally and later operated in a more modern way for more than 19 years in the Jeneberang River with 35 construction materials mining in Gowa Regency [3, 19]. Sand mining activities are practically very beneficial to support the economy of the communities. However, sand mining can lead to excessive exploitation if it is not carried out properly in accordance with conservation principles. Sand miners who do not pay attention to environmental aspects will risk the surrounding area with the danger of erosion and landslides [4].

Beside sand mining activities, sedimentation and the poor management of Bili-Bili Reservoir is also an issue. Bili-Bili Reservoir, located right in the middle of the watershed, was first built in 1992 with the assistance of the Japanese government. It is expected to prevent flooding in the downstream area due to the swift flow of the Jeneberang river from upstream. In 2004, there was massive landslide (167.2 million m^3) occurred in the caldera of Mount Bawakaraeng, causing large amount of sediment carried by the Jeneberang River [5, 16, 24, 25]. Bili-Bili was silted up due to the large amount of landslide sediment carried by the Jeneberang River (75.2 million m^3). In 2009 the total volume of deposits was estimated more than 244.9 million m^3 and unstable sediment deposits which is still exist in the caldera were estimated at 82.7 million m^3, so that the total volume of sediment flowing along the main Jeneberang river channel was 162.2 million m^3 [28]. In 2019 the sedimentation level in the Bili-Bili Reservoir reached a total of 75,055,327 m^3. This sedimentation rate increased very significantly in 2015 reaching an elevation above 60 m [17]. There is already a work plan to carry out dredging in the reservoir, BBWS already has the tools, but the dredging has never been carried out.

Based on our research, we found that sand mining activities are mostly carried out in Tinggimoncong and Parangloe District. According to the Head of Regional Company (*Perusahaan Daerah*), illegal sand mining activities carried out at those areas indirectly damage the artificial embankments (sand pockets). The sand pockets were built to avoid landslides and floods. Sand mining in the riverbanks can destroy both natural and artificial dikes in the Jeneberang River.

Despite its positive impact on regional economic development, a development approach in the context of sand mining will result in market failure due to failure to anticipate environmental impacts or negative externalities of an economic activity. One of the negative externalities of construction material mining comes from mining trucks which can cause conflicts. Our research found that there is contestation of power especially in Parangloe District. Mining activities that are expected to support the local economy of the surrounding communities, in fact, were not beneficials to all parties. The dominance of control over mining points (legal or illegal) carried out by entrepreneurs or individuals has an impact on the welfare of the grassroots community around the Parangloe District and Bili-Bili Reservoir.

WALHI revealed that there were three things that were mutually sustainable because of the exploitation of sand mining activities in the Jeneberang River. First, the silting of the Bili-Bili Reservoir, sand and non-sand material after erosion from mining activities will be carried away by currents and sediment in the Bili-Bili Reservoir. Second, floods and landslides are getting wider because of the silting of

the Bili-Bili Reservoir. The amount of sedimentation in the Bili-Bili Reservoir has decreased its function as a water reservoir. As a result, when the rainfall is high, flooding and landslides are inevitable. And this brings us to the third thing, which is if floods and landslides continue to be unresolved, there will be more casualties and damage. This has been proven in 2019 when floods took many lives in Gowa (53 died), Makassar (1 died), Jeneponto (15 died), and Maros (4 died) [29].

3.4 Downstream: Damages and Its Impacts

The downstream area of Jeneberang Watershed is a densely populated and a rapidly growing urbanized area.[2] But it is still dominated by paddy fields, covering an area of 12,999.46 ha, or 59.3% of the total area. In addition to paddy fields, the downstream area around the riverbank is dominated by built-up areas, especially settlements with an area of 6557.47 ha or 29.9% of the total area. With its diverse land use and outlet of the watershed, the direction of land use zoning in the downstream is permanent cultivation area or 80% utilization area and 20% buffer zone area [30]. In terms of land degradation, 89.09% (3428.05 ha) of the downstream area is classified as *non-degraded*. Meanwhile, the area in *degraded* condition is 302.79 ha or 7.87% of the total downstream area.

Even though most of the areas are classified as *non-degraded*, the downstream area of Jeneberang has become the most vulnerable areas compared to other segments. In 2019, hundreds of houses were flooded in Pangkabinanga Village, Pallangga District, Gowa Regency with an inundation height of about 2 m [2]. This flood was the result of the opening of the floodgates of the Bili-bili Reservoir. A total of 2,121 flood victims living in the riverbank took refuge in several shelters provided by the government [15]. WALHI also recorded that floods and landslides occurred in Gowa, Jeneponto, Maros and Makassar City [29, 32]. The downstream of the watershed also experiences inundation flooding caused by poor drainage. The affected areas include the Tamarunang area [35].

Beside floods, the downstream area becomes more vulnerable due to sedimentation carried by Jeneberang River. As explained in the previous section, sedimentation has not only occurred in the middle stream, but also in the downstream, spreading along the river mouth. The sedimentation[3] that occurred downstream of the Jeneberang River is an example of *cascading risks* caused by the natural processes

[2] Based on our observations, there are many ornamental plant traders who set up kiosk, scavenger activities, chicken farms, food traders as well as plantations activities and illegal gracing, all of which are along the downstream border of the Jeneberang River, which starts from the mouth of Jeneberang River until the rubber dam. All these activities have been contributed to the increase of sedimentation rate and increase watershed pollution. Likewise, fish farming activities in downstream of the Jeneberang River around the Somba Opu Bridge, Gowa Regency.

[3] Some of the sedimentation has created a delta that rises to the surface of the river when the sea water recedes. The delta in some places have been polluted by overgrown of nipah plants (*Nypa fruticans*), water hyacinth plant (*Eichhornia crassipes*), and other weeds.

in the upstream and middle stream areas including the most notable Bawakaraeng's caldera landslide[4] in 2004 [5]. The potential for sediment due to a large avalanche will flow downstream especially in the rainy season, making it prone to floods in the settlements located nearby the river.

Sedimentation in the downstream of Jeneberang is not only caused by natural processes, but also by waste carried by waterways from settlements that flow into the river. This is because certain spots have become waste accumulations in water bodies especially in riverbanks which are close to settlements. According to Yosafat [37], the waste that reaches the river mouth and sea flows from river/streams in the area. Besides, some channels are not optimally functioning. One of them is Jongaya channel in District Tamalate. The same condition has occurred in several drainage areas in each district. About 30% of the existing waterways in the coastal districts of Makassar City (especially Biringkanaya, Tamalanrea, Tallo, and Tamalate are not functioning properly [26]. Beside waste, sedimentation in the downstream is also influenced by fishery activities, cargo ships activities and other activities that have been increasing damage to the riverbanks.

Sedimentation also leads to contamination on the river. Previous report stated that the water quality of the Jeneberang River which has been one of the main raw sources of drinking water for Makassar and its surroundings is quite alarming. The Regional Environment Agency (BLHD) of South Sulawesi classified the water quality of the river crossing Gowa Regency and two districts in Makassar as *very bad*. Jeneberang River water pollution comes from erosion on the riverbanks caused by deforestation, as well as sand mining activities on the banks of the river, so that increasing sedimentation in the river flow. Another problem is anthropogenic activities such as agriculture, plantations, waste disposal, household waste, hotels garbage, and tourism activities.

In Malino and Palangga district for example, Gowa Regency and Tamalate district in Makassar City many housing and markets contribute domestic waste to river bodies. Besides, livestock manure that enters to the river caused getting worse of quality water river [14]. Jeneberang River water is heavily polluted. From observations in the field, not only in rivers, piles and garbage are also found at several points of the Jeneberang embankment in Gowa Regency, because used as garbage dumps by residents around the watershed areas.

The cascading risks downstream also caused by the decreasing function of the water catchment area. The decrease in water catchment areas is caused by rapid urban development, including violations of land use by the community with the constructing buildings on the riverbanks. Many factors cause change in the use of river border areas, including the high rate of urbanization in Makassar City. Population growth in Makassar and its suburban areas is associated with a decrease in agricultural land and an increase in building density [33]. The increasing number of lower-middle income residents has resulted in high demand for settlements, and this has triggered

[4] Nurdin et al. [25] stated that landslides occur due to natural processes, namely there is a disturbance in the stability of the soil making up the slopes which is influenced by geomorphological conditions, especially the slope factor.

the rise of illegal settlements or squatter areas in riverbanks which later affect the risks in Jeneberang Watershed.

3.5 Cascading Risks and Challenges

In the Jeneberang watershed, deforestation has a significant effect on environmental changes in this area. The use of forests that are dominated by human activities, either individually or as a community or as a corporation, and government policies & programs increase the frequency and intensity of hazards, as well as exposure and vulnerability to hazard threats. This condition causes an increase in the danger of landslides and floods due to the reduced/lost function of the forest to absorb rainfall and hold soil and release water periodically. When forest cover is lost, rainwater runoff that carries soil (mud) quickly flows to the lower reaches of the river, thus raising the river level (sediment) and overflowing water causing flooding of agricultural land, surrounding villages, and towns, especially during the rainy season. In contrast, during the dry season, areas downstream of deforestation are vulnerable to drought for months [9, 10].

Agriculture was the main driving factor for forest conversion in the Jeneberang watershed area, but its activities vary between upstream, middle-stream, and downstream areas. In the upper watershed, forest conversion was dominated by two main activities, namely horticultural crop farming and tourism activities. In the middle and downstream areas, deforestation and watershed damage were mainly related to infrastructure development policies and programs. Massive land use change and technological intervention through the construction of the Bili-Bili Reservoir, embankments, and sandbags. These activities were carried out in this watershed area with a steep slope geomorphology.

Vulnerability in the middle and downstream watersheds has increased significantly, resulting in higher systemic and disaster risks, including landslides that have caused high sedimentation in the Jeneberang River and Bili-Bili Reservoir. High sedimentation is a major challenge because it affects the quantity and quality of water in reservoirs which are the source of irrigation water and clean/drinking water. The vulnerability in this area was getting worse due to weather anomalies, especially extreme weather, which tended to increase in frequency. This condition caused an ecological disaster, marked by floods in the rainy season and drought in the dry season, especially the long dry season. Hundreds of families in the downstream area were affected by floods as high as 50 cm that often occurred, especially in the Somba Opu Fort area. On the other hand, during the dry season, the water discharge in the rivers and reservoirs dropped significantly, which often caused drought. Consequently, farmers face a high risk of crop failure due to drought, and residents, especially in downstream and urban areas, including Makassar City, are at high risk of lack of clean and drinking water sources.

Flash flood is an extraordinary event that is characterized by very large floods from higher areas that wash away a lot of material such as water, sand, soil, stones, mud

and wood that move to lower places. This disaster generally begins with a landslide in the upstream river through the slopes along the river. Besides being caused by natural processes, flash floods can also be caused by high deforestation and the event of a dam or reservoir bursting due to heavy rain that flushes the ground for a long time which can result in additional water discharge in the reservoir or dam [6].

4 Conclusions

Communities in watershed areas face high risks stemming from natural hazards and non-natural factors. The Jeneberang watershed area, especially with its geomorphological attributes and high river gradient, is an area that is prone to various types of disasters, especially landslides, floods, and droughts. This condition is exacerbated by extreme weather that is increasingly occurring in this area. Meanwhile, non-natural risks are closely related to structural factors related to development policies and activities in this area. Forest conversion, which was initiated by community agricultural activities, was then encouraged by various tourism sectors, sand mining and policy activities and infrastructure development programs permeated through various economic and service sectors. Economic activities and infrastructure development seen by massive land use changes in every segment (upstream, middlestream, and downstream) have become a systemic trap that results in the destruction of these natural resources. The complexity of these activities has a cascading impact and varies by region in the watershed area. This indicates the high threat and risk of disaster faced by the community starting from the upstream, the middle to the downstream, including in Makassar City.

Cascading risks in the Jeneberang watershed area are marked by the occurrence of several disasters that trigger the emergence of further disasters in the future, considering that one disaster event has the potential to increase the risk of another disaster when community resilience decreases due to continuous disaster intensity. Landslides and floods in the watershed area are examples of the causes of increased vulnerability of the region to disasters.

Poor governance of Jeneberang Watershed, which tend to overlap, also needs to be addressed because it has greatly affected the physical condition of Jeneberang watershed which caused cascading risks in the region. Integration between the central government, local governments, CSOs, and the community is needed to achieve equally just and sustainable watershed management and regional governance. In addition, law enforcement in terms of spatial planning and zoning needs to be done, so that all anthropogenic activities can be carried out by considering land capabilities and in accordance with conservation principles.

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